$f(x,y)=e-(x+y) \qquad 0\le x<\infty, 0\le y<\infty$

Find (a) $P\{X<Y\}$ and (b) $P\{X<a\}$.

6.11. In **Example 1d**, verify that $f(x,y)=2e-xe-2y$, $0<x<\infty$, $0<y<\infty$, is indeed a joint density. function. That is, check that $f(x,y)\ge 0$, and that $\int-\infty\infty\int-\infty\infty f(x,y)dx\,dy=1$.

6.12. The number of people who enter a drugstore in a given hour is a Poisson random variable with parameter $\lambda=10$. Compute the conditional probability that at most 3 men entered the drugstore, given that 10 women entered in that hour. What assumptions have you made?

6.13. A man and a woman agree to meet at a certain location about 12:30 P.M. If the man arrives at a time uniformly distributed between 12:15 and 12:45, and if the woman independently arrives at a time uniformly distributed between 12:00 and 1 P.M., find the probability that the first to arrive waits no longer than 5 minutes. What is the probability that the man arrives first?

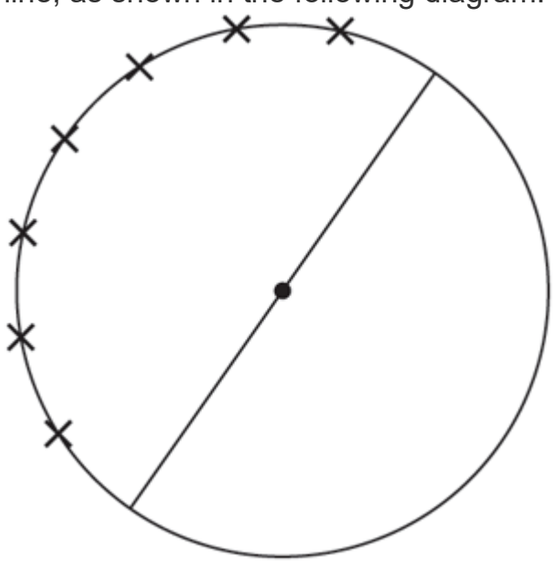
6.14. An ambulance travels back and forth at a constant speed along a road of length L. At a certain moment of time, an accident occurs at a point uniformly distributed on the road. [That is, the distance of the point from one of the fixed ends of the road is uniformly distributed over (0, L).] Assuming that the ambulance's location at the moment of the accident is also uniformly distributed, and assuming independence of the variables, compute the distribution of the distance of the ambulance from the accident.

6.15. The random vector (X, Y) is said to be uniformly distributed over a region R in the plane if, for some constant c, its joint density is

$f(x,y)=\{c \quad if(x, y)\in R0 \quad otherwise$

a. Show that $1/c=$ area of region R.
   Suppose that (X, Y) is uniformly distributed over the square centered at (0, 0) and with sides of length 2.
b. Show that X and Y are independent, with each being distributed uniformly over $(-1,1)$.
c. What is the probability that (X, Y) lies in the circle of radius 1 centered at the origin? That is, find $P\{X2+Y2\le 1\}$.

6.16. Suppose that n points are independently chosen at random on the circumference of a circle, and we want the probability that they all lie in some semicircle. That is, we want the probability that there is a line passing through the center of the circle such that all the points are on one side of that line, as shown in the following diagram:



Let $P1,\ldots,Pn$ denote the n points. Let A denote the event that all the points are contained in some semicircle, and let Ai be the event that all the points lie in the semicircle beginning at the point Pi and going clockwise for $180\circ, i=1,\ldots,n$.

a. Express A in terms of the Ai.

b. Are the Ai mutually exclusive?
c. Find P(A).

6.17. Three points X1,X2,X3 are selected at random on a line L. What is the probability that X2 lies between X1 and X3?
6.18. Let X1 and X2 be independent binomial random variables with Xi having parameters (ni,pi), i=1,2. Find

a. P(X1X2=0);
b. P(X1+X2=1);
c. P(X1+X2=2).

6.19. Show that f(x,y)=1/x, 0<y<x<1, is a joint density function. Assuming that f is the joint density function of X,Y find

a. the marginal density of Y;
b. the marginal density of X;
c. E[X];
d. E[Y].

6.20. The joint density of X and Y is given by
f(x,y)={ x e −(x+y) x>0, y>0 0 otherwise

Are X and Y independent? If, instead, f(x, y) were given by
f(x,y)={ 2 0<x<y, 0<y<1 0 otherwise

would X and Y be independent?
6.21. Let
f(x,y)=24xy 0≤x≤1, 0≤y≤1, 0≤x+y≤1

and let it equal 0 otherwise.

a. Show that f(x y) is a joint probability density function.
b. Find E[X].
c. Find E[Y].

6.22. The joint density function of X and Y is
f(x,y)={ x+y 0<x<1, 0<y<1 0 otherwise

a. Are X and Y independent?
b. Find the density function of X.
c. Find P{X+Y<1}.

6.23. The random variables X and Y have joint density function
f(x,y)=12xy(1−x) 0<x<1, 0<y<1

and equal to 0 otherwise.

a. Are X and Y independent?
b. Find E[X].
c. Find E[Y].
d. Find Var(X).
e. Find Var(Y).

6.24. Consider independent trials, each of which results in outcome i,i=0,1,… ,k with probability pi,∑i=0kpi=1. Let N denote the number of trials needed to obtain an outcome that is not equal to 0,

and let X be that outcome.

a. Find $P\{N=n\}, n \geq 1$.
b. Find $P\{X=j\}, j=1, \ldots, k$.
c. Show that $P\{N=n, X=j\}=P\{N=n\}P\{X=j\}$.
d. Is it intuitive to you that N is independent of X?
e. Is it intuitive to you that X is independent of N?

6.25. Suppose that 106 people arrive at a service station at times that are independent random variables, each of which is uniformly distributed over (0,106). Let N denote the number that arrive in the first hour. Find an approximation for $P\{N=i\}$.

6.26. Suppose that A, B, C, are independent random variables, each being uniformly distributed over (0, 1).

a. What is the joint cumulative distribution function of A, B, C?
b. What is the probability that all of the roots of the equation $Ax^2+Bx+C=0$ are real?

6.27. If X1 and X2 are independent exponential random variables with respective parameters λ_1 and λ_2, find the distribution of $Z=X1/X2$. Also compute $P\{X1<X2\}$.

6.28. The time that it takes to service a car is an exponential random variable with rate 1.

a. If A. J. brings his car in at time 0 and M. J. brings her car in at time t, what is the probability that M. J.'s car is ready before A. J.'s car? (Assume that service times are independent and service begins upon arrival of the car.)
b. If both cars are brought in at time 0, with work starting on M. J.'s car only when A. J.'s car has been completely serviced, what is the probability that M. J.'s car is ready before time 2?

6.29. The gross daily sales at a certain restaurant are a normal random variable with mean $2200 and standard deviation $230. What is the probability that

a. the total gross sales over the next 2 days exceeds $5000;
b. daily sales exceed $2000 in at least 2 of the next 3 days?
   What independence assumptions have you made?

6.30. Jill's bowling scores are approximately normally distributed with mean 170 and standard deviation 20, while Jack's scores are approximately normally distributed with mean 160 and standard deviation 15. If Jack and Jill each bowl one game, then assuming that their scores are independent random variables, approximate the probability that

a. Jack's score is higher;
b. the total of their scores is above 350.

6.31. According to the U.S. National Center for Health Statistics, 25.2 percent of males and 23.6 percent of females never eat breakfast. Suppose that random samples of 200 men and 200 women are chosen. Approximate the probability that

a. at least 110 of these 400 people never eat breakfast;
b. the number of the women who never eat breakfast is at least as large as the number of the men who never eat breakfast.

6.32. Monthly sales are independent normal random variables with mean 100 and standard deviation 5.

a. Find the probability that exactly 3 of the next 6 months have sales greater than 100.
b. Find the probability that the total of the sales in the next 4 months is greater than 420.

6.33. Let X1 and X2 be independent normal random variables, each having mean 10 and variance σ^2. Which probability is larger

a. $P(X1>15)$ or $P(X1+X2>25)$;
b. $P(X1>15)$ or $P(X1+X2>30)$.

6.34. Suppose X and Y are independent normal random variables with mean 10 and variance 4. Find x such that P(X+Y>x)=P(X>15).
6.35. Teams 1,2,3,4 are all scheduled to play each of the other teams 10 times. Whenever team i plays team j, team i is the winner with probability Pi,j, where
P 1,2 =.6, P 1,3 =.7, P 1,4 =.75 P 2,1 =.4, P 2,3 =.6, P 2,4 =.70

a. Approximate the probability that team 1 wins at least 20 games.
   Suppose we want to approximate the probability that team 2 wins at least as many games as does team 1. To do so, let X be the number of games that team 2 wins against team 1, let Y be the total number of games that team 2 wins against teams 3 and 4, and let Z be the total number of games that team 1 wins against teams 3 and 4.
b. Are X,Y,Z independent.
c. Express the event that team 2 wins at least as many games as does team 1 in terms of the random variables X,Y,Z.
d. Approximate the probability that team 2 wins at least as many games as team 1.

Hint: Approximate the distribution of any binomial random variable by a normal with the same mean and variance.
6.36. Let X1,… ,X10 be independent with the same continuous distribution function F, and let m be the median of that distribution. That is, F(m)=.5.

a. If N is the number of the values X1,… ,X10 that are less than m, what type of random variable is N.
b. Let X(1)<X(2)<···<X(10) be the values X1,… ,X10 arranged in increasing order. That is, X(i) is, for i=1, … , 10, the ith smallest of X1,… ,X10. Find P(X(2)<m<X(8)).

6.37. The expected number of typographical errors on a page of a certain magazine is .2. What is the probability that an article of 10 pages contains (a) 0 and (b) 2 or more typographical errors? Explain your reasoning!
6.38. The monthly worldwide average number of airplane crashes of commercial airlines is 2.2. What is the probability that there will be

a. more than 2 such accidents in the next month?
b. more than 4 such accidents in the next 2 months?
c. more than 5 such accidents in the next 3 months?

Explain your reasoning!
6.39. In **Problem 6.4**, calculate the conditional probability mass function of X1 given that

a. X2=1;
b. X2=0.

6.40. In **Problem 6.3**, calculate the conditional probability mass function of Y1 given that

a. Y2=1;
b. Y2=0.

6.41. The discrete integer valued random variables X,Y,Z are independent if for all i,j,k
P(X=i,Y=j,Z=k)=P(X=i)P(Y=j)P(Z=k)

Show that if X,Y,Z are independent then X and Y are independent. That is, show that the preceding implies that
P(X=i,Y=j)=P(X=i)P(Y=j)

6.42. Choose a number X at random from the set of numbers {1,2,3,4,5}. Now choose a number at random from the subset no larger than X, that is, from {1,… ,X}. Call this second number Y.

a. Find the joint mass function of X and Y.
b. Find the conditional mass function of *X* given that Y=i. Do it for i=1,2,3,4,5.
c. Are X and Y independent? Why?

6.43. Two dice are rolled. Let X and Y denote, respectively, the largest and smallest values obtained. Compute the conditional mass function of Y given X=i, for i=1,2,… ,6. Are X and Y independent? Why?

6.44. The joint probability mass function of *X* and Y is given by

p(1,1)= 1 8 p(1,2)= 1 4 p(2,1)= 1 8 p(2,2)= 1 2

a. Compute the conditional mass function of X given Y=i,i=1,2.
b. Are X and Y independent?
c. Compute P{XY≤3},P{X+Y>2},P{X/Y>1}.

6.45. The joint density function of X and Y is given by

f(x,y)=xe−x(y+1) x>0, y>0

a. Find the conditional density of X, given Y=y, and that of Y, given X=x.
b. Find the density function of Z=XY.

6.46. The joint density of X and Y is

f(x,y)=c(x2−y2)e−x 0≤x<∞, −x≤y≤x

Find the conditional distribution of Y, given X=x.

6.47. An insurance company supposes that each person has an accident parameter and that the yearly number of accidents of someone whose accident parameter is λ is Poisson distributed with mean λ. They also suppose that the parameter value of a newly insured person can be assumed to be the value of a gamma random variable with parameters s and α. If a newly insured person has n accidents in her first year, find the conditional density of her accident parameter. Also, determine the expected number of accidents that she will have in the following year.

6.48. If X1,X2,X3 are independent random variables that are uniformly distributed over (0, 1), compute the probability that the largest of the three is greater than the sum of the other two.

6.49. A complex machine is able to operate effectively as long as at least 3 of its 5 motors are functioning. If each motor independently functions for a random amount of time with density function f(x)=xe−x,x>0, compute the density function of the length of time that the machine functions.

6.50. If 3 trucks break down at points randomly distributed on a road of length L, find the probability that no 2 of the trucks are within a distance d of each other when d≤L/2.

6.51. Consider a sample of size 5 from a uniform distribution over (0, 1). Compute the probability that the median is in the interval (14,34).

6.52. If X1,X2,X3,X4,X5 are independent and identically distributed exponential random variables with the parameter λ, compute

a. P{min(X1,… ,X5)≤a};
b. P{max(X1,… ,X5)≤a}.

6.53. Let X(1),X(2),…,X(n) be the order statistics of a set of n independent uniform (0,1) random variables. Find the conditional distribution of X(n) given that X(1)=s1,X(2)=s2,… ,X(n−1)=sn−1.

6.54. Let Z1 and Z2 be independent standard normal random variables. Show that X,Y has a bivariate normal distribution when X=Z1, Y=Z1+Z2.

6.55. Derive the distribution of the range of a sample of size 2 from a distribution having density function f(x)=2x, 0<x<1.

6.56. Let X and Y denote the coordinates of a point uniformly chosen in the circle of radius 1 centered at the origin. That is, their joint density is

f(x,y)=1π x2+y2≤1

Find the joint density function of the polar coordinates R=(X2+Y2)1/2 and Θ=tan−1Y/X.
6.57. If X and Y are independent random variables both uniformly distributed over (0, 1), find the joint density function of R=X2+Y2,Θ=tan−1Y/X.
6.58. If U is uniform on (0, 2π) and Z, independent of U, is exponential with rate 1, show directly (without using the results of **Example 7b**) that X and Y defined by
X= 2Z cos U Y= 2Z sin U

are independent standard normal random variables.
6.59. X and Y have joint density function
f(x,y)=1x2y2 x≥1, y≥1

a. Compute the joint density function of U=XY,V=X/Y.
b. What are the marginal densities?

6.60. If X and Y are independent and identically distributed uniform random variables on (0, 1), compute the joint density of
a. U=X+Y,V=X/Y;
b. U=X,V=X/Y;
c. U=X+Y,V=X/(X+Y).

6.61. Repeat **Problem 6.60** when X and Y are independent exponential random variables, each with parameter λ=1.
6.62. If X1 and X2 are independent exponential random variables, each having parameter λ, find the joint density function of Y1=X1+X2 and Y2=eX1.
6.63. If X, Y, and Z are independent random variables having identical density functions f(x)=e−x,0<x<∞, derive the joint distribution of U=X+Y, V=X+Z,W=Y+ Z.
6.64. In Example 8b, let Yk+1=n+1−∑i=1kYi. Show that Y1,… ,Yk,Yk+1 are exchangeable. Note that Yk+1 is the number of balls one must observe to obtain a special ball if one considers the balls in their reverse order of withdrawal.
6.65. Consider an urn containing n balls numbered 1,… ,n, and suppose that k of them are randomly withdrawn. Let Xi equal 1 if ball number i is removed and let Xi be 0 otherwise. Show that X1,… ,Xn are exchangeable.

Theoretical exercises

6.1. Suppose X, Y have a joint distribution function F(x,y). Show how to obtain the distribution functions FX(x)=P{X≤x} and FY(y)=P{Y≤y}.
6.2. Suppose that X and Y are integer valued random variables and have a joint distribution function F(i,j)=P(X≤i,Y≤j).

a. Give an expression, in terms of the joint distribution function, for P(X=i,Y≤j).
b. Give an expression, in terms of the joint distribution function, for P(X=i,Y=j).

6.3. Suggest a procedure for using Buffon's needle problem to estimate π. Surprisingly enough, this was once a common method of evaluating π.
6.4. Solve Buffon's needle problem when L>D.
ANSWER: 2LπD(1−sin θ)+2θ/π, where cos θ=D/L.
6.5. If X and Y are independent continuous positive random variables, express the density function of (a) Z=X/Y and (b) Z=XY in terms of the density functions of X and Y. Evaluate the density functions in the special case where X and Y are both exponential random variables.
6.6. If X and Y are jointly continuous with joint density function fX,Y(x,y), show that X+Y is continuous with density function
fX+Y(t)=∫−∞∞fX,Y(x,t−x) dx

6.7.

a. If X has a gamma distribution with parameters (t,λ), what is the distribution of cX,c>0?
b. Show that
   12λχ2n2

   has a gamma distribution with parameters n,λ when n is a positive integer and χ22n is a chi-squared random variable with 2n degrees of freedom.

6.8. Let X and Y be independent continuous random variables with respective hazard rate functions λX(t) and λY(t), and set W=min(X,Y).

a. Determine the distribution function of W in terms of those of X and Y.
b. Show that λW(t), the hazard rate function of W, is given by
   λW(t)=λX(t)+λY(t)

6.9. Let X1,… ,Xn be independent exponential random variables having a common parameter λ. Determine the distribution of min(X1,… ,Xn).
6.10. The lifetimes of batteries are independent exponential random variables, each having parameter λ. A flashlight needs 2 batteries to work. If one has a flashlight and a stockpile of n batteries, what is the distribution of time that the flashlight can operate?
6.11. Let X1, X2, X3, X4, X5 be independent continuous random variables having a common distribution function F and density function f, and set
I=P{X1<X2<X3<X4<X5}

a. Show that I does not depend on F.
   Hint: Write I as a five-dimensional integral and make the change of variables ui=F(xi),i=1,… ,5.
b. Evaluate I.
c. Give an intuitive explanation for your answer to (b).

6.12. Show that the jointly continuous (discrete) random variables X1,… ,Xn are independent if and only if their joint probability density (mass) function f(x1,… ,xn) can be written as
f(x1,… ,xn)=∏i=1ngi(xi)

for nonnegative functions $g_i(x), i=1,\ldots,n$.

6.13. In **Example 5e**, we computed the conditional density of a success probability for a sequence of trials when the first $n+m$ trials resulted in n successes. Would the conditional density change if we specified which n of these trials resulted in successes?

6.14. Suppose that X and Y are independent geometric random variables with the same parameter p.

a. Without any computations, what do you think is the value of $P\{X=i|X+Y=n\}$?

   Hint: Imagine that you continually flip a coin having probability p of coming up heads. If the second head occurs on the nth flip, what is the probability mass function of the time of the first head?

b. Verify your conjecture in part (a).

6.15. Consider a sequence of independent trials, with each trial being a success with probability p. Given that the kth success occurs on trial n, show that all possible outcomes of the first $n-1$ trials that consist of $k-1$ successes and $n-k$ failures are equally likely.

6.16. If X and Y are independent binomial random variables with identical parameters n and p, show analytically that the conditional distribution of X given that $X+Y=m$ is the hypergeometric distribution. Also, give a second argument that yields the same result without any computations.

Hint: Suppose that $2n$ coins are flipped. Let X denote the number of heads in the first n flips and Y the number in the second n flips. Argue that given a total of m heads, the number of heads in the first n flips has the same distribution as the number of white balls selected when a sample of size m is chosen from n white and n black balls.

6.17. Suppose that $X_i, i=1,2,3$ are independent Poisson random variables with respective means $\lambda_i, i=1,2,3$. Let $X=X_1+X_2$ and $Y=X_2+X_3$. The random vector X,Y is said to have a bivariate Poisson distribution. Find its joint probability mass function. That is, find $P\{X=n, Y=m\}$.

6.18. Suppose X and Y are both integer-valued random variables. Let

$$p(i|j)=P(X=i|Y=j)$$

and

$$q(j|i)=P(Y=j|X=i)$$

Show that

$$P(X=i,Y=j)= p(i|j) \sum_i p(i|j)\, q(j|i)$$

6.19. Let X_1, X_2, X_3 be independent and identically distributed continuous random variables. Compute

a. $P\{X_1>X_2|X_1>X_3\}$;
b. $P\{X_1>X_2|X_1<X_3\}$;
c. $P\{X_1>X_2|X_2>X_3\}$;
d. $P\{X_1>X_2|X_2<X_3\}$.

6.20. Let U denote a random variable uniformly distributed over $(0, 1)$. Compute the conditional distribution of U given that

a. $U>a$;
b. $U<a$;

where $0<a<1$.

6.21. Suppose that W, the amount of moisture in the air on a given day, is a gamma random variable with parameters (t,β). That is, its density is $f(w)=\beta e^{-\beta w}(\beta w)^{t-1}/\Gamma(t)$, $w>0$. Suppose also that given that $W=w$, the number of accidents during that day—call it N—has a Poisson distribution with mean w. Show that the conditional distribution of W given that $N=n$ is the gamma distribution with parameters $(t+n,\beta+1)$.

6.22. Let W be a gamma random variable with parameters (t,β), and suppose that conditional on

W=w,X1,X2,… ,Xn are independent exponential random variables with rate w. Show that the conditional distribution of W given that X1=x1,X2=x2,… ,Xn=xn is gamma with parameters (t+n,β+∑i=1nxi).

6.23. A rectangular array of *mn* numbers arranged in n rows, each consisting of m columns, is said to contain a *saddlepoint* if there is a number that is both the minimum of its row and the maximum of its column. For instance, in the array

1 3 2 0 −2 6 .5 12 3

the number 1 in the first row, first column is a saddlepoint. The existence of a saddlepoint is of significance in the theory of games. Consider a rectangular array of numbers as described previously and suppose that there are two individuals—A and B—who are playing the following game: A is to choose one of the numbers 1,2,… ,n and B one of the numbers 1,2,… ,m. These choices are announced simultaneously, and if A chose i and B chose j, then A wins from B the amount specified by the number in the ith row, jth column of the array. Now suppose that the array contains a saddlepoint—say the number in the row r and column k call this number xrk. Now if player A chooses row r, then that player can guarantee herself a win of at least xrk (since xrk is the minimum number in the row r). On the other hand, if player B chooses column k, then he can guarantee that he will lose no more than xrk (since xrk is the maximum number in the column k). Hence, as A has a way of playing that guarantees her a win of xrk and as B has a way of playing that guarantees he will lose no more than xrk, it seems reasonable to take these two strategies as being optimal and declare that the value of the game to player A is xrk.

If the *nm* numbers in the rectangular array described are independently chosen from an arbitrary continuous distribution, what is the probability that the resulting array will contain a saddlepoint?

6.24. If X is exponential with rate λ, find P{[X]=n,X−[X]≤x}, where [x] is defined as the largest integer less than or equal to x. Can you conclude that [X] and X−[X] are independent?

6.25. Suppose that F(x) is a cumulative distribution function. Show that (a) Fn(x) and (b) 1−[1−F(x)]n are also cumulative distribution functions when n is a positive integer.

Hint: Let X1,… ,Xn be independent random variables having the common distribution function F. Define random variables Y and Z in terms of the Xi so that P{Y≤x}=Fn(x) and P{Z≤x}=1−[1−F(x)]n.

6.26. Show that if n people are distributed at random along a road L miles long, then the probability that no 2 people are less than a distance D miles apart is when D≤L/(n−1),[1−(n−1)D/L]n. What if D>L/(n−1)?

6.27. Suppose that X1,… ,Xn are independent exponential random variables with rate λ. Find

a. fX1|X1+… +Xn(x|t), the conditional density of X1 given that X1+… +Xn=t;
b. P(X1<x|X1+… +Xn=t).

6.28. Establish **Equation (6.2)** by differentiating **Equation (6.4)**.

6.29. Show that the median of a sample of size 2n+1 from a uniform distribution on (0, 1) has a beta distribution with parameters (n+1, n+1).

6.30. Suppose that X1,… ,Xn are independent and identically distributed continuous random variables. For A={X1<⋯<Xj>Xj+1>⋯>Xn}, find P(A). That is, find the probability that the function X(i)=Xi, i=1,… ,n, is a unimodal function with maximal value X(j). **Hint:** Write

$$A=\{\max(X_1,\ldots,X_j)=\max(X_1,\ldots,X_n),\ X_1<\cdots<X_j,\ X_{j+1}>\cdots>X_n\}$$

6.31. Compute the density of the range of a sample of size n from a continuous distribution having density function f.

6.32. Let X(1)≤X(2)≤⋯≤X(n) be the ordered values of n independent uniform (0, 1) random variables. Prove that for 1≤k≤n+1,

P{X(k)−X(k−1)>t}=(1−t)n

where X(0)≡0,X(n+1)≡1, and 0<t<1.

6.33. Let X1,… ,Xn be a set of independent and identically distributed continuous random variables having distribution function F, and let X(i),i=1,… ,n denote their ordered values. If X, independent of the Xi,i=1,… ,n, also has distribution F, determine

a. P{X>X(n)};
b. P{X>X(1)};
c. P{X(i)<X<X(j)},1≤i<j≤n.

6.34. Let X1,… ,Xn be independent and identically distributed random variables having distribution function F and density f. The quantity M≡[X(1)+X(n)]/2, defined to be the average of the smallest and largest values in X1,… ,Xn, is called the *midrange* of the sequence. Show that its distribution function is
FM(m)=n∫−∞m[F(2m−x)−F(x)]n−1f(x) dx

6.35. Let X1,… ,Xn be independent uniform (0, 1) random variables. Let R=X(n)−X(1) denote the range and M=[X(n)+X(1)]/2 the midrange of X1,… ,Xn. Compute the joint density function of R and M.
6.36. If X and Y are independent standard normal random variables, determine the joint density function of
U=X V=XY

Then use your result to show that X/Y has a Cauchy distribution.
6.37. Suppose that (X,Y) has a bivariate normal distribution with parameters μx,μy,σx,σy,ρ.
a. Show that (X− μ x σx , Y− μ y σy) has a bivariate normal distribution with parameters 0,1,0,1,ρ.
b. What is the joint distribution of (aX+b,cY+d).

6.38. Suppose that X has a beta distribution with parameters (a,b), and that the conditional distribution of N given that X=x is binomial with parameters (n+m,x). Show that the conditional density of X given that N=n is the density of a beta random variable with parameters (n+a,m+b). N is said to be a *beta binomial* random variable.
6.39. Consider an experiment with n possible outcomes, having respective probabilities P1,… ,Pn, ∑i=1nPi=1, and suppose we want to assume a probability distribution on the probability vector (P1,… ,Pn). Because ∑i=1nPi=1, we cannot define a density on P1,… ,Pn, but what we can do is to define one on P1,… ,Pn−1 and then take Pn=1−∑i=1n−1Pi. The *Dirichlet* distribution takes (P1,… ,Pn−1) to be uniformly distributed over the set S={(p1,… ,pn−1):∑i=1n−1pi<1, pi>0, i=1,… ,n−1}. That is, the Dirichlet density is
f P 1 ,… , P n−1 (p 1 ,… , p n−1)=C, p i >0,i=1,… ,n−1, ∑ i=1 n−1 p i <1

a. Determine C. **Hint**: Use results from **Section 6.3.1**.
Let U1,… ,Un be independent uniform (0,1) random variables.
b. Show that the Dirichlet density is the conditional density of U1,… ,Un−1 given that ∑i=1n−1Ui<1.
c. * Show that U(1),U(2)−U(1),… ,U(n)−U(n−1) has a Dirichlet distribution, where U(1),… ,U(n) are the order statistics of U1,… ,Un.

6.40. Let FX1,… ,Xn(x1,… ,xn) and fX1,… ,Xn(x1,… ,xn) be, respectively, the joint distribution function and the joint density function of X1,… ,Xn.
Show that
∂n∂x1⋯∂xnFX1,… ,Xn(x1,… ,xn)=fX1,… ,Xn(x1,… ,xn).

6.41. For given constants ci>0, let Yi=ciXi, i=1,… ,n, and let FY1,… ,Yn(x1,… ,xn) and fY1,… ,Yn(x1,… ,xn) be, respectively the joint distribution function and the joint density function of Y1,… ,Yn
a. Express FY1,… ,Yn(x1,… ,xn) in terms of the joint distribution function of X1,… ,Xn
b. Express fY1,… ,Yn(x1,… ,xn) in terms of the joint density function of X1,… ,Xn.
c. Use **Equation (7.3)** to verify your answer to part (b).

Self-Test Problems and Exercises

6.1. Each throw of an unfair die lands on each of the odd numbers 1, 3, 5 with probability C and on each of the even numbers with probability 2C.

a. Find C.
b. Suppose that the die is tossed. Let X equal 1 if the result is an even number, and let it be 0 otherwise. Also, let Y equal 1 if the result is a number greater than three and let it be 0 otherwise. Find the joint probability mass function of X and Y. Suppose now that 12 independent tosses of the die are made.
c. Find the probability that each of the six outcomes occurs exactly twice.
d. Find the probability that 4 of the outcomes are either one or two, 4 are either three or four, and 4 are either five or six.
e. Find the probability that at least 8 of the tosses land on even numbers.

6.2. The joint probability mass function of the random variables X, Y, Z is
p(1,2,3)=p(2,1,1)=p(2,2,1)=p(2,3,2)=14

Find (a) E[XYZ], and (b) E[XY+XZ+YZ].
6.3. The joint density of X and Y is given by
f(x,y)=C(y−x)e−y −y<x<y, 0<y<∞

a. Find C.
b. Find the density function of X.
c. Find the density function of Y.
d. Find E[X].
e. Find E[Y].

6.4. Let r=r1+… +rk, where all ri are positive integers. Argue that if X1,… ,Xr has a multinomial distribution, then so does Y1,… ,Yk where, with r0=0,
Yi=∑j=ri−1+1ri−1+riXj , i≤k

That is, Y1 is the sum of the first r1 of the X′s, Y2 is the sum of the next r2, and so on.
6.5. Suppose that X, Y, and Z are independent random variables that are each equally likely to be either 1 or 2. Find the probability mass function of (a) XYZ, (b) XY+XZ+YZ, and (c) X2+YZ.
6.6. Let X and Y be continuous random variables with joint density function
f(x,y)={ x 5 +cy 0<x<1, 1<y<5 0 otherwise

where c is a constant.

a. What is the value of c?
b. Are X and Y independent?
c. Find P{X+Y>3}.

6.7. The joint density function of X and Y is
f(x,y)={ xy 0<x<1, 0<y<2 0 otherwise

a. Are X and Y independent?
b. Find the density function of X.
c. Find the density function of Y.
d. Find the joint distribution function.
e. Find E[Y].
f. Find P{X+Y<1}.

6.8. Consider two components and three types of shocks. A type 1 shock causes component 1 to fail, a type 2 shock causes component 2 to fail, and a type 3 shock causes both components 1 and 2 to fail. The times until shocks 1, 2, and 3 occur are independent exponential random variables with respective rates $\lambda 1, \lambda 2$, and $\lambda 3$. Let Xi denote the time at which component i fails, i=1,2. The random variables X1,X2 are said to have a joint bivariate exponential distribution. Find $P\{X1>s, X2>t\}$.

6.9. Consider a directory of classified advertisements that consists of m pages, where m is very large. Suppose that the number of advertisements per page varies and that your only method of finding out how many advertisements there are on a specified page is to count them. In addition, suppose that there are too many pages for it to be feasible to make a complete count of the total number of advertisements and that your objective is to choose a directory advertisement in such a way that each of them has an equal chance of being selected.

a. If you randomly choose a page and then randomly choose an advertisement from that page, would that satisfy your objective? Why or why not?
   Let n(i) denote the number of advertisements on page i,i=1,… ,m, and suppose that whereas these quantities are unknown, we can assume that they are all less than or equal to some specified value n. Consider the following algorithm for choosing an advertisement.
   Step 1Choose a page at random. Suppose it is page X. Determine n(X) by counting the number of advertisements on page X.
   Step 2"Accept" page X with probability n(X)/n. If page X is accepted, go to step 3. Otherwise, return to step 1.
   Step 3Randomly choose one of the advertisements on page X.

   Call each pass of the algorithm through step 1 an iteration. For instance, if the first randomly chosen page is rejected and the second accepted, then we would have needed 2 iterations of the algorithm to obtain an advertisement.
b. What is the probability that a single iteration of the algorithm results in the acceptance of an advertisement on page i?
c. What is the probability that a single iteration of the algorithm results in the acceptance of an advertisement?
d. What is the probability that the algorithm goes through k iterations, accepting the jth advertisement on page i on the final iteration?
e. What is the probability that the jth advertisement on page i is the advertisement obtained from the algorithm?
f. What is the expected number of iterations taken by the algorithm?

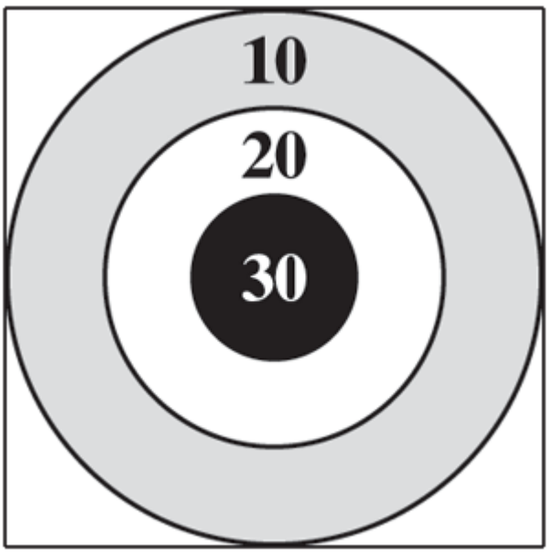
6.10. The "random" parts of the algorithm in Self-Test **Problem 6.9** can be written in terms of the generated values of a sequence of independent uniform (0, 1) random variables, known as random numbers. With [x] defined as the largest integer less than or equal to x, the first step can be written as follows:
Step 1Generate a uniform (0, 1) random variable U. Let X=[mU]+1, and determine the value of n(X).

a. Explain why the above is equivalent to step 1 of **Problem 6.8**.
   ***Hint*:** What is the probability mass function of X?
b. Write the remaining steps of the algorithm in a similar style.

6.11. Let X1,X2,… be a sequence of independent uniform (0, 1) random variables. For a fixed constant c, define the random variable N by
N=min{n:Xn>c}

Is N independent of XN? That is, does knowing the value of the first random variable that is greater than c affect the probability distribution of when this random variable occurs? Give an intuitive explanation for your answer.

6.12. The accompanying dartboard is a square whose sides are of length 6:



The three circles are all centered at the center of the board and are of radii 1, 2, and 3, respectively. Darts landing within the circle of radius 1 score 30 points, those landing outside this circle, but within the circle of radius 2, are worth 20 points, and those landing outside the circle of radius 2, but within the circle of radius 3, are worth 10 points. Darts that do not land within the circle of radius 3 do not score any points. Assuming that each dart that you throw will, independently of what occurred on your previous throws, land on a point uniformly distributed in the square, find the probabilities of the accompanying events:

a. You score 20 on a throw of the dart.
b. You score at least 20 on a throw of the dart.
c. You score 0 on a throw of the dart.
d. The expected value of your score on a throw of the dart.
e. Both of your first two throws score at least 10.
f. Your total score after two throws is 30.

6.13. A model proposed for NBA basketball supposes that when two teams with roughly the same record play each other, the number of points scored in a quarter by the home team minus the number scored by the visiting team is approximately a normal random variable with mean 1.5 and variance 6. In addition, the model supposes that the point differentials for the four quarters are independent. Assume that this model is correct.

a. What is the probability that the home team wins?
b. What is the conditional probability that the home team wins, given that it is behind by 5 points at halftime?
c. What is the conditional probability that the home team wins, given that it is ahead by 5 points at the end of the first quarter?

6.14. Let N be a geometric random variable with parameter p. Suppose that the conditional distribution of X given that N=n is the gamma distribution with parameters n and λ. Find the conditional probability mass function of N given that X=x.

6.15. Let X and Y be independent uniform (0, 1) random variables.

a. Find the joint density of U=X,V=X+Y.
b. Use the result obtained in part (a) to compute the density function of V.

6.16. You and three other people are to place bids for an object, with the high bid winning. If you win, you plan to sell the object immediately for $10,000. How much should you bid to maximize your expected profit if you believe that the bids of the others can be regarded as being independent and uniformly distributed between $7,000 and $10,000 thousand dollars?

6.17. Find the probability that X1,X2,… ,Xn is a permutation of 1,2,… ,n, when X1,X2,… ,Xn are

independent and

a. each is equally likely to be any of the values 1,… , n;
b. each has the probability mass function P{Xi=j}=pj,j=1,… , n.

6.18. Let X1,… ,Xn and Y1,… ,Yn be independent random vectors, with each vector being a random ordering of k ones and n−k zeros. That is, their joint probability mass functions are
P{ X 1 = i 1 ,… , X n = i n } =P{ Y 1 = i 1 ,… , Y n = i n } = 1 (n k) , i j =0,1, ∑ j=1 n i j =k

Let
N=∑i=1n|Xi−Yi|

denote the number of coordinates at which the two vectors have different values. Also, let M denote the number of values of i for which Xi=1,Yi=0.

a. Relate N to M.
b. What is the distribution of M?
c. Find E[N].
d. Find Var(N).

*6.19. Let Z1,Z2,… ,Zn be independent standard normal random variables, and let
Sj=∑i=1jZi

a. What is the conditional distribution of Sn given that Sk=y. Find it for k=1,… ,n−1.
b. Show that, for 1≤k≤n, the conditional distribution of Sk given that Sn=x is normal with mean xk/n and variance k(n−k)/n.

6.20. Let X1,X2,… be a sequence of independent and identically distributed continuous random variables. Find

a. P{X6>X1|X1=max(X1,… ,X5)}
b. P{X6>X2|X1=max(X1,… ,X5)}

6.21. Prove the identity
P{ X≤s,Y≤t }=P{ X≤s }+P{ Y≤t }+P{X>s,Y>t}−1

Hint: Derive an expression for P{X>s,Y>t} by taking the probability of the complementary event.
6.22. In **Example 1c**, find P(Xr=i,Ys=j) when j<i.
6.23. A Pareto random variable X with parameters a>0, >0 has distribution function F(x)=1− a λ x −λ ,x>a. x>a. For x0>a, verify that the conditional distribution of X given that X>x0 is that of a Pareto random variable with parameters (x0,λ) by evaluating P(X>x|X>x0).
6.24. Verify the identity fX(x)=∫−∞∞fX|Y(x|y)fY(y)dy.
6.25. In a contest originating with n players, each player independently advances to the next round, with player i advancing with probability pi. If no players advance to the next round, then the contest ends and all the players in the just concluded round are declared co-winners. If only one player advances, then that player is declared the winner and the contest ends. If two or more players advance, then those players play another round. Let Xi denote the number of rounds that i plays.

a. Find P(Xi≥k). **Hint:** Note that {Xi≥k} will occur if i advances at least k times and at least one of the other players advances at least k−1 times.
b. Find *P*(*i* is either the sole winner or one of the co-winners).
   Hint: It might help to imagine that a player always continues to play rounds until he or she fails to advance. (That is, if there is a sole winner then imagine that that player continues on until a failure occurs.)
c. Find *P*(*i* is the sole winner)

6.26. Let X1,… ,Xn be independent nonnegative integer valued random variables, and let

Suppose that we are presented with a set of n distinct values x1,x2,…,xn and that we desire to put them in increasing order, or as it is commonly stated, to *sort* them. An efficient procedure for accomplishing this task is the quick-sort algorithm, which is defined as follows: When n=2, the algorithm compares the two values and then puts them in the appropriate order. When n>2, one of the elements is randomly chosen say it is xi and then all of the other values are compared with xi. Those smaller than xi are put in a bracket to the left of xi and those larger than xi are put in a bracket to the right of xi. The algorithm then repeats itself on these brackets and continues until all values have been sorted. For instance, suppose that we desire to sort the following 10 distinct values:

5,9,3,10,11,14,8,4,17,6

We start by choosing one of them at random (that is, each value has probability 110 of being chosen). Suppose, for instance, that the value 10 is chosen. We then compare each of the others to this value, putting in a bracket to the left of 10 all those values smaller than 10 and to the right all those larger. This gives

{5,9,3,8,4,6},10,{11,14,17}

We now focus on a bracketed set that contains more than a single value say–the one on the left of the preceding–and randomly choose one of its values–say that 6 is chosen. Comparing each of the values in the bracket with 6 and putting the smaller ones in a new bracket to the left of 6 and the larger ones in a bracket to the right of 6 gives

{5,3,4},6,{9,8},10,{11,14,17}

If we now consider the leftmost bracket, and randomly choose the value 4 for comparison, then the next iteration yields

{3},4,{5},6,{9,8},10,{11,14,17}

This continues until there is no bracketed set that contains more than a single value.

If we let X denote the number of comparisons that it takes the quick-sort algorithm to sort n distinct numbers, then E[X] is a measure of the effectiveness of this algorithm. To compute E[X], we will first express X as a sum of other random variables as follows. To begin, give the following names to the values that are to be sorted: Let 1 stand for the smallest, let 2 stand for the next smallest, and so on. Then, for $1 \le i < j \le n$, let I(i,j) equal 1 if i and j are ever directly compared, and let it equal 0 otherwise. With this definition, it follows that

$$X=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} I(i,j)$$

implying that

$$E[X]=E\left[\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} I(i,j)\right]=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} E[I(i,j)]=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n} P\{i \text{ and } j \text{ are ever compared}\}$$

To determine the probability that i and j are ever compared, note that the values i,i+1,…,j−1,j will initially be in the same bracket (since all values are initially in the same bracket) and will remain in the same bracket if the number chosen for the first comparison is not between i and j. For instance, if the comparison number is larger than j, then all the values i,i+1,…,j−1,j will go in a bracket to the left of the comparison number, and if it is smaller than i, then they will all go in a bracket to the right. Thus all the values i,i+1,…,j−1,j will remain in the same bracket until the first time that one of them is chosen as a comparison value. At that point all the other values between i and j will be compared with this comparison value. Now, if this comparison value is neither i nor j, then upon comparison with it, i will go into a left bracket and j into a right bracket, and thus i and j will be in different brackets and so will never be compared. On the other hand, if the comparison value of the set i,i+1,…,j−1,j is either i or j, then there will be a direct comparison between i and j. Now, given that the comparison value is one of the values between i and j, it follows that it is equally likely to be any of these j−i+1 values, and thus the probability that it is either i or j is 2/(j−i+1). Therefore, we can conclude that

$$P\{i \text{ and } j \text{ are ever compared}\}=2j-i+1$$

and

$$E[X]=\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}2j-i+1$$

To obtain a rough approximation of the magnitude of E[X] when n is large, we can approximate the sums by integrals. Now

$$\sum_{j=i+1}^{n}2j-i+1\approx\int_{i+1}^{n}2x-i+1\,dx=2\log(x-i+1)\Big|_{i+1}^{n}=2\log(n-i+1)-2\log(2)\approx 2\log(n-i+1)$$

Thus

$$E[X]\approx\sum_{i=1}^{n-1}2\log(n-i+1)\approx 2\int_{1}^{n-1}\log(n-x+1)\,dx=2\int_{2}^{n}\log(y)\,dy=2(y\log(y)-y)\Big|_{2}^{n}\approx 2n\log(n)$$

Thus we see that when n is large, the quick-sort algorithm requires, on average, approximately 2n log(n) comparisons to sort n distinct values.

Example 2n The probability of a union of events

Let $A_1,\ldots A_n$ denote events, and define the indicator variables $X_i, i=1,\ldots,n$, by

$$X_i=\begin{cases}1 & \text{if } A_i \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

Now, note that

$$1-\prod_{i=1}^{n}(1-X_i)=\begin{cases}1 & \text{if } \cup A_i \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

Hence,

$$E\left[1-\prod_{i=1}^{n}(1-X_i)\right]=P(\cup_{i=1}^{n}A_i)$$

Expanding the left side of the preceding formula yields

$$P(\cup_{i=1}^{n}A_i)=E\left[\sum_{i=1}^{n}X_i-\sum\sum_{i<j}X_iX_j+\sum\sum\sum_{i<j<k}X_iX_jX_k-\cdots+(-1)^{n+1}X_1\cdots X_n\right] \tag{2.3}$$

However,

$$X_{i_1}X_{i_2}\cdots X_{i_k}=\begin{cases}1 & \text{if } A_{i_1}A_{i_2}\cdots A_{i_k} \text{ occurs}\\ 0 & \text{otherwise}\end{cases}$$

so

$$E[X_{i_1}\cdots X_{i_k}]=P(A_{i_1}\cdots A_{i_k})$$

Thus, **Equation (2.3)** is just a statement of the well-known inclusion-exclusion formula for the union of events:

$$P(\cup A_i)=\sum_i P(A_i)-\sum\sum_{i<j}P(A_iA_j)+\sum\sum\sum_{i<j<k}P(A_iA_jA_k)-\cdots+(-1)^{n+1}P(A_1\cdots A_n)$$

When one is dealing with an infinite collection of random variables $X_i, i\geq 1$, each having a finite expectation, it is not necessarily true that

$$E\left[\sum_{i=1}^{\infty}X_i\right]=\sum_{i=1}^{\infty}E[X_i] \tag{2.4}$$

To determine when (**2.4**) is valid, we note that $\sum_{i=1}^{\infty}X_i=\lim_{n\to\infty}\sum_{i=1}^{n}X_i$. Thus,

$$E\left[\sum_{i=1}^{\infty} X_i\right] = E\left[\lim_{n\to\infty} \sum_{i=1}^{n} X_i\right] = ? \lim_{n\to\infty} E\left[\sum_{i=1}^{n} X_i\right] = \lim_{n\to\infty} \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{\infty} E[X_i] \quad (2.5)$$

Hence, **Equation (2.4)** is valid whenever we are justified in interchanging the expectation and limit operations in **Equation (2.5)**. Although, in general, this interchange is *not* justified, it can be shown to be valid in two important special cases:

1. The Xi are all nonnegative random variables. (That is, P{Xi≥0}=1 for all i.)
2. ∑i=1∞E[|Xi|]<∞.

Example 2o

Consider any nonnegative, integer-valued random variable X. If, for each i≥1, we define

$$X_i = \begin{cases} 1 & \text{if } X \geq i \\ 0 & \text{if } X < i \end{cases}$$

then

$$\sum_{i=1}^{\infty} X_i = \sum_{i=1}^{X} X_i + \sum_{i=X+1}^{\infty} X_i = \sum_{i=1}^{X} 1 + \sum_{i=X+1}^{\infty} 0 = X$$

Hence, since the Xi are all nonnegative, we obtain

$$E[X] = \sum_{i=1}^{\infty} E(X_i) = \sum_{i=1}^{\infty} P\{X \geq i\} \quad (2.6)$$

a useful identity.

Example 2p

Suppose that n elements—call them 1, 2, …, n—must be stored in a computer in the form of an ordered list. Each unit of time, a request will be made for one of these elements —i being requested, independently of the past, with probability P(i), i≥1, ∑iP(i)=1. Assuming that these probabilities are known, what ordering minimizes the average position in the line of the element requested?

Solution

Suppose that the elements are numbered so that P(1)≥P(2)≥···≥P(n). To show that 1, 2, …, n is the optimal ordering, let X denote the position of the requested element. Now, under any ordering say, O=i1,i2,…,in,

$$P_O\{X \geq k\} = \sum_{j=k}^{n} P(i_j) \geq \sum_{j=k}^{n} P(j) = P_{1,2,\ldots,n}\{X \geq k\}$$

Summing over k and using **Equation (2.6)** yields

$$E_O[X] \geq E_{1,2,\ldots,n}[X]$$

thus showing that ordering the elements in decreasing order of the probability that they are requested minimizes the expected position of the element requested.

*7.2.1 Obtaining bounds from expectations via the probabilistic method

The probabilistic method is a technique for analyzing the properties of the elements of a set by introducing probabilities on the set and then studying an element chosen according to those probabilities. The technique was previously seen in Example 4l of **Chapter 3**, where it was used to show that a set contained an element that satisfied a certain property. In this subsection, we show how it can sometimes be used to bound complicated functions.

Let f be a function on the elements of a finite set A, and suppose that we are interested in

$m=\max_{s\in A} f(s)$

A useful lower bound for m can often be obtained by letting S be a random element of A for which the expected value of f(S) is computable and then noting that $m\geq f(S)$ implies that

$m\geq E[f(S)]$

with strict inequality if f(S) is not a constant random variable. That is, E[f(S)] is a lower bound on the maximum value.

Example 2q The maximum number of hamiltonian paths in a tournament

A round-robin tournament of $n>2$ contestants is a tournament in which each of the $\binom{n}{2}$ pairs of contestants play each other exactly once. Suppose that the players are numbered 1,2,3,…,n. The permutation $i_1,i_2,\ldots i_n$ is said to be a *Hamiltonian path* if i_1 beats i_2, i_2 beats i_3,…, and i_{n-1} beats i_n. A problem of some interest is to determine the largest possible number of Hamiltonian paths.

As an illustration, suppose that there are 3 players. On the one hand, one of them wins twice, then there is a single Hamiltonian path. (For instance, if 1 wins twice and 2 beats 3, then the only Hamiltonian path is 1, 2, 3.) On the other hand, if each of the players wins once, then there are 3 Hamiltonian paths. (For instance, if 1 beats 2, 2 beats 3, and 3 beats 1, then 1, 2, 3; 2, 3, 1; and 3, 1, 2, are all Hamiltonians.) Hence, when n=3, there is a maximum of 3 Hamiltonian paths.

We now show that there is an outcome of the tournament that results in more than $n!/2^{n-1}$ Hamiltonian paths. To begin, let the outcome of the tournament specify the result of each of the $\binom{n}{2}$ games played, and let A denote the set of all $2^{\binom{n}{2}}$ possible tournament outcomes. Then, with f(s) defined as the number of Hamiltonian paths that result when the outcome is $s\in A$, we are asked to show that

$\max_s f(s)\geq n!2^{n-1}$

To show this, consider the randomly chosen outcome S that is obtained when the results of the $\binom{n}{2}$ games are independent, with each contestant being equally likely to win each encounter. To determine E[f(S)], the expected number of Hamiltonian paths that result from the outcome S, number the n! permutations, and, for i=1,…,n!, let

$$X_i=\begin{cases}1 & \text{if permutation } i \text{ is a Hamiltonian}\\ 0 & \text{otherwise}\end{cases}$$

Since

$f(S)=\sum_i X_i$

it follows that

$E[f(S)]=\sum_i E[X_i]$

Because, by the assumed independence of the outcomes of the games, the probability that any specified permutation is a Hamiltonian is $(1/2)^{n-1}$, it follows that

$E[X_i]=P\{X_i=1\}=(1/2)^{n-1}$

Therefore,

$E[f(S)]=n!(1/2)^{n-1}$

Since f(S) is not a constant random variable, the preceding equation implies that there is an outcome of the tournament having more than $n!/2^{n-1}$ Hamiltonian paths.

Example 2r

A grove of 52 trees is arranged in a circular fashion. If 15 chipmunks live in these trees, show that there is a group of 7 consecutive trees that together house at least 3 chipmunks.

Solution

Let the neighborhood of a tree consist of that tree along with the next six trees visited by moving in the clockwise direction. We want to show that for any choice of living accommodations of the 15 chipmunks, there is a tree that has at least 3 chipmunks living in its neighborhood. To show this, choose a tree at random and let X denote the number of chipmunks that live in its neighborhood. To determine E[X], arbitrarily number the 15 chipmunks and for i=1,…,15, let

$$X_i=\begin{cases} 1, & \text{if chipmunk } i \text{ lives in the neighborhood of the randomly chosen tree} \\ 0, & \text{otherwise} \end{cases}$$

Because

$$X=\sum_{i=1}^{15}X_i$$

we obtain that

$$E[X]=\sum_{i=1}^{15}E[X_i]$$

However, because Xi will equal 1 if the randomly chosen tree is any of the 7 trees consisting of the tree in which chipmunk i lives along with its 6 neighboring trees when moving in the counterclockwise direction,

$$E[X_i]=P\{X_i=1\}=\frac{7}{52}$$

Consequently,

$$E[X]=\frac{105}{52}>2$$

showing that there exists a tree with more than 2 chipmunks living in its neighborhood.

*7.2.2 The maximum–minimums identity

We start with an identity relating the maximum of a set of numbers to the minimums of the subsets of these numbers.

Proposition 2.2

For arbitrary numbers $x_i, i=1,\ldots,n$,

$$\max_i x_i=\sum_i x_i-\sum_{i<j}\min(x_i,x_j)+\sum_{i<j<k}\min(x_i,x_j,x_k)$$

$$+\ldots+(-1)^{n+1}\min(x_1,\ldots,x_n)$$

Proof We will give a probabilistic proof of the proposition. To begin, assume that all the x_i are in the interval [0, 1]. Let U be a uniform (0, 1) random variable, and define the events $A_i, i=1,\ldots,n$, by $A_i=\{U<x_i\}$. That is, A_i is the event that the uniform random variable is less than x_i. Because at least one of these events A_i will occur if U is less than at least one of the values x_i, we have that

$$\cup_i A_i=\{U<\max_i x_i\}$$

Therefore,

$P(\cup_i A_i) = P\{U < \max_i x_i\} = \max_i x_i$

Also,

$P(A_i) = P\{U < x_i\} = x_i$

In addition, because all of the events $A_{i1}, \ldots, A_{ir}$ will occur if U is less than all the values $x_{i1}, \ldots, x_{ir}$, we see that the intersection of these events is

$A_{i1} \ldots A_{ir} = \{U < \min_{j=1,\ldots r} x_{ij}\}$

implying that

$P(A_{i1} \ldots A_{ir}) = P\{U < \min_{j=1,\ldots r} x_{ij}\} = \min_{j=1,\ldots r} x_{ij}$

Thus, the proposition follows from the inclusion–exclusion formula for the probability of the union of events:

$$P(\cup_i A_i) = \sum_i P(A_i) - \sum_{i<j} P(A_i A_j) + \sum_{i<j<k} P(A_i A_j A_k) + \ldots + (-1)^{n+1} P(A_1 \ldots A_n)$$

When the x_i are nonnegative, but not restricted to the unit interval, let c be such that all the x_i are less than c. Then the identity holds for the values $y_i = x_i/c$, and the desired result follows by multiplying through by c. When the x_i can be negative, let b be such that $x_i + b > 0$ for all i. Therefore, by the preceding,

$$\max_i (x_i + b) = \sum_i (x_i + b) - \sum_{i<j} \min(x_i + b, x_j + b) + \cdots + (-1)^{n+1} \min(x_1 + b, \ldots, x_n + b)$$

Letting

$$M = \sum_i x_i - \sum_{i<j} \min(x_i, x_j) + \cdots + (-1)^{n+1} \min(x_1, \ldots, x_n)$$

we can rewrite the foregoing identity as

$$\max_i x_i + b = M + b\left(n - \binom{n}{2} + \cdots + (-1)^{n+1} \binom{n}{n}\right)$$

But

$$0 = (1-1)^n = 1 - n + \binom{n}{2} + \cdots + (-1)^n \binom{n}{n}$$

The preceding two equations show that

$$\max_i x_i = M$$

and the proposition is proven.

It follows from **Proposition 2.2** that for any random variables $X_1, \ldots, X_n$,

$$\max_i X_i = \sum_i X_i - \sum_{i<j} \min(X_i, X_j) + \cdots + (-1)^{n+1} \min(X_1, \ldots, X_n)$$

Taking expectations of both sides of this equality yields the following relationship between the expected value of the maximum and those of the partial minimums:

$$E[\max_i X_i] = \sum_i E[X_i] - \sum_{i<j} E[\min(X_i, X_j)] + \cdots + (-1)^{n+1} E[\min(X_1, \ldots, X_n)] \quad (2.7)$$

Example 2s Coupon collecting with unequal probabilities

Suppose there are n types of coupons and that each time one collects a coupon, it is, independently of previous coupons collected, a type i coupon with probability p_i, $\sum_{i=1}^n p_i = 1$. Find the expected number of

coupons one needs to collect to obtain a complete set of at least one of each type.

Solution

If we let Xi denote the number of coupons one needs to collect to obtain a type i, then we can express X as

$$X=\max_{i=1,\ldots,n} X_i$$

Because each new coupon obtained is a type i with probability pi, Xi is a geometric random variable with parameter pi. Also, because the minimum of Xi and Xj is the number of coupons needed to obtain either a type i or a type j, it follows that for $i \neq j$, min (Xi,Xj) is a geometric random variable with parameter pi+pj. Similarly, min (Xi,Xj,Xk), the number needed to obtain any of types i, j, or k, is a geometric random variable with parameter pi+pj+pk, and so on. Therefore, the identity (**2.7**) yields

$$E[X]=\sum_{i=1} p_i-\sum_{i<j} p_i+p_j+\sum_{i<j<k} p_i+p_j+p_k+\cdots+(-1)^{n+1} p_1+\cdots+p_n$$

Noting that

$$\int_0^\infty e^{-px}\,dx=\frac{1}{p}$$

and using the identity

$$1-\prod_{i=1}^n (1-e^{-p_i x})=\sum_i e^{-p_i x}-\sum_{i<j} e^{-(p_i+p_j)x}+\cdots+(-1)^{n+1} e^{-(p_1+\cdots+p_n)x}$$

shows, upon integrating the identity, that

$$E[X]=\int_0^\infty \left(1-\prod_{i=1}^n (1-e^{-p_i x})\right)\,dx$$

which is a more useful computational form.

7.3 Moments of the Number of Events that Occur

Many of the examples solved in the previous section were of the following form: For given events A1,…,An, find E[X], where X is the number of these events that occur. The solution then involved defining an indicator variable Ii for event Ai such that

Ii={1, if Ai occurs0, otherwise

Because

X=∑i=1nIi

we obtained the result

E[X]=E[∑i=1nIi]=∑i=1nE[Ii]=∑i=1nP(Ai) (3.1)

Now suppose we are interested in the number of *pairs* of events that occur. Because IiIj will equal 1 if both Ai and Aj occur and will equal 0 otherwise, it follows that the number of pairs is equal to ∑i<jIiIj. But because X is the number of events that occur, it also follows that the number of pairs of events that occur is (X2). Consequently,

(X2)=∑i<jIiIj

where there are (n2) terms in the summation. Taking expectations yields

E[(X2)]=∑i<j E[IiIj]=∑i<jP(AiAj) (3.2)

or

E[X(X−1)2]=∑i<jP(AiAj)

giving that

E[X2]−E[X]=2∑i<jP(AiAj) (3.3)

which yields $E[X^2]$, and thus Var (X)=E[X2]−(E[X])2.

Moreover, by considering the number of distinct subsets of k events that all occur, we see that

(Xk)=∑i1<i2<…<ikIi1Ii2⋯Iik

Taking expectations gives the identity

E[(Xk)]=∑i1<i2<…<ikE[Ii1Ii2⋯Iik]=∑i1<i2<…<ikP(Ai1Ai2⋯Aik) (3.4)

Example 3a Moments of binomial random variables

Consider n independent trials, with each trial being a success with probability p . Let A_i be the event that trial i is a success. When i=j, P(AiAj)=p2. Consequently, **Equation (3.2**) yields

E[(X2)]=∑i<jp2=(n2)p2

or

$$E[X(X-1)]=n(n-1)p2$$

or

$$E[X2]-E[X]=n(n-1)p2$$

Now, $E[X]=\sum_{i=1}^{n}P(Ai)=np$, so, from the preceding equation

$$\text{Var}(X)=E[X2]-(E[X])2=n(n-1)p2+np-(np)2=np(1-p)$$

which is in agreement with the result obtained in **Section 4.6.1**.

In general, because $P(Ai1Ai2\cdots Aik)=pk$, we obtain from **Equation (3.4)** that

$$E[(Xk)]=\sum_{i1<i2<\ldots<ik}pk=(nk)pk$$

or, equivalently,

$$E[X(X-1)\cdots(X-k+1)]=n(n-1)\cdots(n-k+1)pk$$

The successive values $E[Xk]$, $k\geq 3$, can be recursively obtained from this identity. For instance, with $k=3$, it yields

$$E[X(X-1)(X-2)]=n(n-1)(n-2)p3$$

or

$$E[X3-3X2+2X]=n(n-1)(n-2)p3$$

or

$$E[\,X3\,]=3E[\,X2\,]-2E[X]+n(n-1)(n-2)p3=3n(n-1)p2+np+n(n-1)(n-2)p3$$

Example 3b Moments of hypergeometric random variables

Suppose n balls are randomly selected from an urn containing N balls, of which m are white. Let Ai be the event that the ith ball selected is white. Then X, the number of white balls selected, is equal to the number of the events A1,…,An that occur. Because the ith ball selected is equally likely to be any of the N balls, of which m are white, $P(Ai)=m/N$. Consequently, **Equation (3.1)** gives that $E[X]=\sum_{i=1}^{n}P(Ai)=nm/N$. Also, since

$$P(AiAj)=P(Ai)P(Aj|Ai)=mNm-1N-1$$

we obtain, from **Equation (3.2)**, that

$$E[(X2)]=\sum_{i<j}m(m-1)N(N-1)=(n2)m(m-1)N(N-1)$$

or

$$E[X(X-1)]=n(n-1)m(m-1)N(N-1)$$

showing that

$$E[X2]=n(n-1)m(m-1)N(N-1)+E[X]$$

This formula yields the variance of the hypergeometric, namely,

$$Var\ (X)=E[\ X2\ ]-(E[X])2=n(n-1)m(m-1)N(N-1)+nmN-n2m2N2=mnN[\ (n-1)(m-1)N-1+1-mnN\ ]$$

which agrees with the result obtained in **Example 8j** of **Chapter 4**.

Higher moments of X are obtained by using **Equation (3.4)**. Because

$$P(Ai1Ai2\cdots Aik)=m(m-1)\cdots(m-k+1)N(N-1)\cdots(N-k+1)$$

Equation (3.4) yields

$$E[(Xk)]=(nk)m(m-1)\cdots(m-k+1)N(N-1)\cdots(N-k+1)$$

or

$$E[\ X(X-1)\cdots(X-k+1\ )]\qquad =n(n-1)\cdots(n-k+1)m(m-1)\cdots(m-k+1)N(N-1)\cdots(N-k+1)$$

Example 3c Moments in the match problem

For i=1,…,N, let Ai be the event that person i selects his or her own hat in the match problem. Then

$$P(AiAj)=P(Ai)P(Aj|Ai)=1N1N-1$$

which follows because, conditional on person i selecting her own hat, the hat selected by person j is equally likely to be any of the other N−1 hats, of which one is his own. Consequently, with X equal to the number of people who select their own hat, it follows from **Equation (3.2)** that

$$E[(X2)]=\sum i<j1N(N-1)=(N2)1N(N-1)$$

thus showing that

$$E[X(X-1)]=1$$

Therefore, $E[\ X2\ ]=1+E[\ X\ ]$. Because $E[\ X\ ]=\sum i=1NP(Ai) = 1$, we obtain that

$$Var\ (X)=E[X2]-(E[X])2=1.$$

Hence, both the mean and variance of the number of matches is 1. For higher moments, we use **Equation (3.4)**, along with the fact that $P(Ai1Ai2\cdots Aik)=1N(N-1)\cdots(N-k+1)$, to obtain

$$E[\ (X\ k)\ ]=(N\ k)1N(N-1)\cdots(N-k+1)$$

or

$$E[X(X-1)\cdots(X-k+1)]=1$$

Example 3d Another coupon-collecting problem

Suppose that there are N distinct types of coupons and that, independently of past types collected, each new one obtained is type j with probability pj, $\sum j=1Npj=1$. Find the expected value and variance of the

number of different types of coupons that appear among the first n collected.

Solution

We will find it more convenient to work with the number of uncollected types. So, let Y equal the number of types of coupons collected, and let X=N−Y denote the number of uncollected types. With Ai defined as the event that there are no type i coupons in the collection, X is equal to the number of the events A1,…,AN that occur. Because the types of the successive coupons collected are independent, and, with probability 1−pi each new coupon is not type i, we have

$$P(A_i)=(1-p_i)^n$$

Hence, $E[X]=\sum_{i=1}^{N}(1-p_i)^n$, from which it follows that

$$E[Y]=N-E[X]=N-\sum_{i=1}^{N}(1-p_i)^n$$

Similarly, because each of the n coupons collected is neither a type i nor a type j coupon, with probability 1−pi−pj, we have

$$P(A_iA_j)=(1-p_i-p_j)^n, \quad i\neq j$$

Thus,

$$E[X(X-1)]=2\sum_{i<j}P(A_iA_j)=2\sum_{i<j}(1-p_i-p_j)^n$$

or

$$E[X^2]=2\sum_{i<j}(1-p_i-p_j)^n+E[X]$$

Hence, we obtain

$$\text{Var}(Y)=\text{Var}(X)=E[X^2]-(E[X])^2=2\sum_{i<j}(1-p_i-p_j)^n+\sum_{i=1}^{N}(1-p_i)^n-\left(\sum_{i=1}^{N}(1-p_i)^n\right)^2$$

In the special case where $p_i=1/N$, $i=1,\ldots,N$, the preceding formulas give

$$E[Y]=N[1-(1-1N)^n]$$

and

$$\text{Var}(Y)=N(N-1)(1-2N)^n+N(1-1N)^n-N^2(1-1N)^{2n}$$

Example 3e The negative hypergeometric random variables

Suppose an urn contains n+m balls, of which n are special and m are ordinary. These items are removed one at a time, with each new removal being equally likely to be any of the balls that remain in the urn. The random variable Y, equal to the number of balls that need be withdrawn until a total of r special balls have been removed, is said to have a *negative hypergeometric distribution*. The negative hypergeometric distribution bears the same relationship to the hypergeometric distribution as the negative binomial does to the binomial. That is, in both cases, rather than considering a random variable equal to the number of successes in a fixed number of trials (as are the binomial and hypergeometric variables), they refer to the number of trials needed to obtain a fixed number of successes.

To obtain the probability mass function of a negative hypergeometric random variable Y, note that Y will

equal k if both

1. the first k−1 withdrawals consist of r−1 special and k−r ordinary balls and
2. the kth ball withdrawn is special.

Consequently,

$$P\{ Y=k \}=(\ nr-1)(\ mk-r)(n+mk-1)n-r+1n+m-k+1$$

We will not, however, utilize the preceding probability mass function to obtain the mean and variance of Y. Rather, let us number the m ordinary balls as o1,…,om, and then, for each i=1,…,n, let Ai be the event that oi is withdrawn before r special balls have been removed. Then, if X is the number of the events A1,…,Am that occur, it follows that X is the number of ordinary balls that are withdrawn before a total of r special balls have been removed. Consequently,

$$Y=r+X$$

showing that

$$E[Y]=r+E[X]=r+\sum i=1mP(Ai)$$

To determine P(Ai), consider the n+1 balls consisting of oi along with the n special balls. Of these n+1 balls, oi is equally likely to be the first one withdrawn, or the second one withdrawn, …, or the final one withdrawn. Hence, the probability that it is among the first r of these to be selected (and so is removed before a total or r special balls have been withdrawn) is rn+1. Consequently,

$$P(Ai)=rn+1$$

and

$$E[Y]=r+mrn+1=r(n+m+1)n+1$$

Thus, for instance, the expected number of cards of a well-shuffled deck that would need to be turned over until a spade appears is 1+3914=3.786, and the expected number of cards that would need to be turned over until an ace appears is 1+485=10.6.

To determine $Var(Y)=Var(X)$, we use the identity

$$E[X(X-1)]=2\sum i<jP(AiAj)$$

Now, P(AiAj) is the probability that both oi and oj are removed before there have been a total of r special balls removed. So consider the n+2 balls consisting of oi,oj, and the n special balls. Because all withdrawal orderings of these balls are equally likely, the probability that oi and oj are both among the first r+1 of them to be removed (and so are both removed before r special balls have been withdrawn) is

$$P(AiAj)=(22)(\ nr-1)(n+2r+1)=r(r+1)(n+1)(n+2)$$

Consequently,

$$E[X(X-1)]=2(m2)r(r+1)(n+1)(n+2)$$

so

$$E[X2]=m(m-1)r(r+1)(n+1)(n+2)+E[X]$$

Because E[X]=mrn+1, this yields

$$\mathrm{Var}(Y)=\mathrm{Var}(X)=m(m-1)r(r+1)(n+1)(n+2)+mrn+1-(mrn+1)2$$

A little algebra now shows that

$$\mathrm{Var}(Y)=mr(n+1-r)(n+m+1)(n+1)2(n+2)$$

Example 3f Singletons in the coupon collector's problem

Suppose that there are n distinct types of coupons and that, independently of past types collected, each new one obtained is equally likely to be any of the n types. Suppose also that one continues to collect coupons until a complete set of at least one of each type has been obtained. Find the expected value and variance of the number of types for which exactly one coupon of that type is collected.

Solution

Let X equal the number of types for which exactly one of that type is collected. Also, let Ti denote the ith type of coupon to be collected, and let Ai be the event that there is only a single type Ti coupon in the complete set. Because X is equal to the number of the events A1,…,An that occur, we have

$$E[X]=\sum_{i=1}^{n}P(A_i)$$

Now, at the moment when the first type Ti coupon is collected, there remain n−i types that need to be collected to have a complete set. Because, starting at this moment, each of these n−i+1 types (the $n-i$ not yet collected and type T_i) is equally likely to be the last of these types to be collected, it follows that the type T_i will be the last of these types (and so will be a singleton) with probability 1n−i+1. Consequently, P(Ai)=1n−i+1, yielding

$$E[X]=\sum_{i=1}^{n}\frac{1}{n-i+1}=\sum_{i=1}^{n}\frac{1}{i}$$

To determine the variance of the number of singletons, let Si, j, for i<j, be the event that the first type Ti coupon to be collected is still the only one of its type to have been collected at the moment that the first type Tj coupon has been collected. Then

$$P(A_iA_j)=P(A_iA_j|S_{i,j})P(S_{i,j})$$

Now, P(Si,j) is the probability that when a type Ti has just been collected, of the n−i+1 types consisting of type Ti and the n−i as yet uncollected types, a type Ti is not among the first j−i of these types to be collected. Because type Ti is equally likely to be the first, or second, or …,n−i+1 of these types to be collected, we have

$$P(S_{i,j})=\frac{1-j-i}{n-i+1}=\frac{n+1-j}{n+1-i}$$

Now, conditional on the event Si,j, both Ai and Aj will occur if, at the time the first type Tj coupon is collected, of the n−j+2 types consisting of types Ti,Tj, and the n−j as yet uncollected types, Ti and Tj are both collected after the other n−j. But this implies that

$$P(A_iA_j|S_{i,j})=\frac{2}{n-j+2}\frac{1}{n-j+1}$$

Therefore,

$$P(A_iA_j)=\frac{2}{(n+1-i)(n+2-j)},\quad i<j$$

yielding

$$E[X(X-1)]=4\sum_{i<j}^{1}(n+1-i)(n+2-j)$$

Consequently, using the previous result for E[X], we obtain

$$\text{Var}(X)=4\sum_{i<j}^{1}(n+1-i)(n+2-j)+\sum_{i=1}^{n}1i-(\sum_{i=1}^{n}1i)2$$

7.4 Covariance, Variance of Sums, and Correlations

The following proposition shows that the expectation of a product of independent random variables is equal to the product of their expectations.

Proposition 4.1

If X and Y are independent, then, for any functions h and g,

$$E[g(X)h(Y)]=E[g(X)]E[h(Y)]$$

Proof Suppose that X and Y are jointly continuous with joint density f(x,y). Then

$$E[g(x)h(Y)]=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}g(x)\,h(y)\,f(x,y)\,dx\,dy=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}g(x)\,h(y)\,f_X(x)\,f_Y(y)\,dx\,dy=\int_{-\infty}^{\infty}h(y)\,f_Y(y)dy\int_{-\infty}^{\infty}g(x)\,f_X(x)\,dx=E[h(Y)]E[g(X)]$$

The proof in the discrete case is similar.

Just as the expected value and the variance of a single random variable give us information about that random variable, so does the covariance between two random variables give us information about the relationship between the random variables.

> Definition
> The covariance between X and Y, denoted by Cov (X,Y), is defined by
>
> $$Cov(X,Y)=E[(X-E[X])(Y-E[Y])]$$

Upon expanding the right side of the preceding definition, we see that

$$Cov\,(X,Y)=E[XY-E[X]Y-XE[Y]+E[Y]E[X]]=E[XY]-E[X]E[Y]-E[X]E[Y]+E[X]E[Y]=E[XY]-E[X]E[Y]$$

Note that if X and Y are independent, then, by **Proposition 4.1**, Cov (X,Y)=0. However, the converse is not true. A simple example of two dependent random variables X and Y having zero covariance is obtained by letting X be a random variable such that

$$P\{X=0\}=P\{X=1\}=P\{X=-1\}=13$$

and defining

$$Y=\begin{cases}0 & \text{if } X\neq 0\\ 1 & \text{if } X=0\end{cases}$$

Now,XY=0, so E[XY]=0. Also,E[X]=0. Thus,

$$Cov\,(X,Y)=E[XY]-E[X]E[Y]=0$$

However,X and Y are clearly not independent.

The following proposition lists some of the properties of covariance.

Proposition 4.2

i. Cov (X,Y)= Cov (Y,X)
ii. Cov (X,X)=Cov(X)
iii. Cov (aX,Y)=a Cov (X,Y)
iv. $Cov\ (\sum_{i=1}^{n}X_i, \sum_{j=1}^{m}Y_j)=\sum_{i=1}^{n}\sum_{j=1}^{m} Cov\,(X_i,Y_j)$

Proof of Proposition 4.2 Parts (i) and (ii) follow immediately from the definition of covariance, and part (iii) is left as an exercise for the reader. To prove part (iv), which states that the covariance operation is additive (as is the operation of taking expectations), let $\mu_i=E[X_i]$ and $v_j=E[Y_j]$. Then

$$E[\sum_{i=1}^{n}X_i]=\sum_{i=1}^{n}\mu_i,\quad E[\sum_{j=1}^{m}Y_j]=\sum_{j=1}^{m}v_j$$

and

$$Cov\,(\sum_{i=1}^{n}X_i, \sum_{j=1}^{m}Y_j)=E\,[\,(\sum_{i=1}^{n}X_i-\sum_{i=1}^{n}\mu_i)(\sum_{j=1}^{m}Y_j-\sum_{j=1}^{m}v_j)\,]=E\,[\,\sum_{i=1}^{n}(X_i-\mu_i)-\sum_{j=1}^{m}(Y_j-v_j)\,]=E\,[\,\sum_{i=1}^{n}\sum_{j=1}^{m}(X_i-\mu_i)(Y_j-v_j)]=\sum_{i=1}^{n}\sum_{j=1}^{m}E\,[\,(X_i-\mu_i\,)(Y_j-v_j)]$$

where the last equality follows because the expected value of a sum of random variables is equal to the sum of the expected values.

It follows from parts (ii) and (iv) of **Proposition 4.2**, upon taking Yj=Xj,j=1,…,n, that

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right)=\text{Cov}\left(\sum_{i=1}^{n}X_i,\sum_{j=1}^{n}X_j\right)=\sum_{i=1}^{n}\sum_{j=1}^{n}\text{Cov}(X_i,X_j)=\sum_{i=1}^{n}\left(\text{Cov}(X_i,X_i)+\sum_{j:j\neq i}\text{Cov}(X_i,X_j)\right)=\sum_{i=1}^{n}\text{Var}(X_i)+\sum\sum_{i\neq j}\text{Cov}(X_i,X_j)$$

Since each pair of indices i,j,i≠j, appears twice in the double summation, the preceding formula is equivalent to

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right)=\sum_{i=1}^{n}\text{Var}(X_i)+2\sum\sum_{i<j}\text{Cov}(X_i,X_j) \qquad (4.1)$$

If X1,…,Xn are pairwise independent, in that Xi and Xj are independent for i≠j, then **Equation (4.1)** reduces to

$$\text{Var}\left(\sum_{i=1}^{n}X_i\right)=\sum_{i=1}^{n}\text{Var}(X_i)$$

The following examples illustrate the use of **Equation (4.1)**.

Example 4a

Let X1,…,Xn be independent and identically distributed random variables having expected value μ and variance σ2, and as in **Example 2c**, let $\bar{X}=\sum_{i=1}^{n}X_i/n$ be the sample mean. The quantities $X_i-\bar{X}$,i=1,…,n, are called *deviations*, as they equal the differences between the individual data and the sample mean. The random variable

$$S^2=\sum_{i=1}^{n}(X_i-\bar{X})^2/n-1$$

is called the *sample variance*. Find (a) $\text{Var}(\bar{X})$ and (b)E[S2].

Solution

a. $\text{Var}(\bar{X})=(1/n)^2\,\text{Var}\left(\sum_{i=1}^{n}X_i\right)=(1/n)^2\sum_{i=1}^{n}\text{Var}(X_i)$ by independence$=\sigma^2/n$

b. We start with the following algebraic identity:
$$(n-1)S^2=\sum_{i=1}^{n}(X_i-\mu+\mu-\bar{X})^2=\sum_{i=1}^{n}(X_i-\mu)^2+\sum_{i=1}^{n}(\bar{X}-\mu)^2-2(\bar{X}-\mu)\sum_{i=1}^{n}(X_i-\mu)=\sum_{i=1}^{n}(X_i-\mu)^2+n(\bar{X}-\mu)^2-2(\bar{X}-\mu)n(\bar{X}-\mu)$$
$$=\sum_{i=1}^{n}(X_i-\mu)^2-n(\bar{X}-\mu)^2$$

Taking expectations of the preceding yields

$$(n-1)E[S^2]=\sum_{i=1}^{n}E[(X_i-\mu)^2]-nE[(\bar{X}-\mu)^2]=n\sigma^2-n\,\text{Var}(\bar{X})=(n-1)\sigma^2$$

where the final equality made use of part (a) of this example and the one preceding it made use of the result of **Example 2c**, namely, that $E[\bar{X}]=\mu$. Dividing through by n−1 shows that the expected value of the sample variance is the distribution variance σ2.

Our next example presents another method for obtaining the variance of a binomial random variable.

Example 4b Variance of a binomial random variable

Compute the variance of a binomial random variable X with parameters n and p.

Solution

Since such a random variable represents the number of successes in n independent trials when each trial has the common probability p of being a success, we may write

$$X=X_1+\cdots+X_n$$

where the Xi are independent Bernoulli random variables such that

$$X_i=\begin{cases}1 & \text{if the } i\text{th trial is a success}\\ 0 & \text{otherwise}\end{cases}$$

Hence, from **Equation (4.1)**, we obtain

$$\text{Var}(X)=\text{Var}(X_1)+\cdots+\text{Var}(X_n)$$